```
> # In this worksheet we compute the Matrix associated with MM-
  Kinects for a given number of Substrates and Enzymes
  restart;with(LinearAlgebra):with(Student):

> # First, you have to be sure that you are using the latest version
  of Physics Package, otherwise some commands may not work properly.
  In this work, we used the update Physics61.3 availabe on
  http://www.maplesoft.com/products/maple/features/physicsresearch.
  aspx
  Physics:-Version();
"/Library/Frameworks/Maple.framework/Versions/17/lib/Physics.mla",                (1)
   2014, March 19, 3:0 hours

> interface(rtablesize=51);
                                    10                                           (2)

> Mat:=proc(NS,NE,kp1,km1,k2)


  delta:=proc(a,b) piecewise(a=b,1,0); end proc;
  Baseset:=proc(sm,em,cm,pm) global B; global Bdim;
  i:=1;

  for si from 0 to sm
      do
         for ei from 0 to em
             do
                for ci from 0 to cm
                    do
                       for pri from 0 to pm
                           do
                           if ei+ci = em then
                           if si+ci+pri = sm then

                           B[i]:=(si,ei,ci,pri):
                           i:=i+1;
                           end if;
                           end if;
                           end do;
                    end do;
                end do;
          end do;
      Bdim:=i-1;
  end proc;
  Baseset(NS,NE,NE,NS):

  ME:=proc(si,sit,ei,eit,ci,cit,pri,prit,kp1,km1,k2) (-delta(ei, eit)
  *delta(ci, cit)*((k2+km1)*cit+kp1*sit*eit)*delta(si, sit)+kp1*sit*
  eit*delta(si, sit-1)*delta(ei, eit-1)*delta(ci, cit+1)+km1*cit*
  delta(si, sit+1)*delta(ei, eit+1)*delta(ci, cit-1))*delta(pri,
  prit)+k2*cit*delta(si, sit)*delta(ei, eit+1)*delta(ci, cit-1)*delta
  (pri, prit+1); end proc;

  element:= proc(i,j,kp1,km1,k2) ME(B[i][1],B[j][1],B[i][2],B[j][2],B
  [i][3],B[j][3],B[i][4],B[j][4],kp1,km1,k2); end proc;
```

```
    for i from 1 to Bdim do
        for j from 1 to Bdim do
        A[i,j]:= element(i,j,kp1,km1,k2);
        end do;
    end do;
  Mt:= Array(1..Bdim,1..Bdim,(i,j)-> A[i,j]);
  end proc;
```

```
Warning, `delta` is implicitly declared local to procedure `Mat`
Warning, `Baseset` is implicitly declared local to procedure `Mat`
Warning, `ME` is implicitly declared local to procedure `Mat`
Warning, `element` is implicitly declared local to procedure `Mat`
Warning, `i` is implicitly declared local to procedure `Mat`
Warning, `j` is implicitly declared local to procedure `Mat`
Warning, `A` is implicitly declared local to procedure `Mat`
Warning, `Mt` is implicitly declared local to procedure `Mat`
Warning, `si` is implicitly declared local to procedure `Baseset`
Warning, `ei` is implicitly declared local to procedure `Baseset`
Warning, `ci` is implicitly declared local to procedure `Baseset`
Warning, `pri` is implicitly declared local to procedure `Baseset`
```

$Mat := \mathbf{proc}(NS, NE, kp1, km1, k2)$ **(3)**

local δ, *Baseset*, *ME*, *element*, *i*, *j*, *A*, *Mt*;

$\delta := \mathbf{proc}(a, b)\ piecewise(a = b, 1, 0)\ \mathbf{end\ proc}$;

$Baseset := \mathbf{proc}(sm, em, cm, pm)$

local *si*, *ei*, *ci*, *pri*;

global *B*, *Bdim*;

$i := 1$;

for *si* **from** 0 **to** *sm* **do**

for *ei* **from** 0 **to** *em* **do**

for *ci* **from** 0 **to** *cm* **do**

for *pri* **from** 0 **to** *pm* **do**

if $ei + ci = em$ **then**

if $si + ci + pri = sm$ **then** $B[i] := si, ei, ci, pri;\ i := i + 1$ **end if**

end if

end do

end do

end do

end do;

$Bdim := i - 1$

end proc;

$Baseset(NS, NE, NE, NS)$;

$ME := \mathbf{proc}(si, sit, ei, eit, ci, cit, pri, prit, kp1, km1, k2)$

$(-\delta(ei, eit) * \delta(ci, cit) * ((k2 + km1) * cit + kp1 * sit * eit) * \delta(si, sit) + kp1 * sit * eit$

```
        * δ(si, sit − 1) * δ(ei, eit − 1) * δ(ci, cit + 1) + km1 * cit * δ(si, sit + 1) * δ(ei, eit
        + 1) * δ(ci, cit − 1)) * δ(pri, prit) + k2 * cit * δ(si, sit) * δ(ei, eit + 1) * δ(ci,
        cit − 1) * δ(pri, prit + 1)
    end proc;
    element := proc(i, j, kp1, km1, k2)
        ME(B[i][1], B[j][1], B[i][2], B[j][2], B[i][3], B[j][3], B[i][4], B[j][4], kp1, km1,
        k2)
    end proc;
    for i to Bdim do for j to Bdim do A[i, j] := element(i, j, kp1, km1, k2) end do end do;
    Mt := Array(1 ..Bdim, 1 ..Bdim, (i, j) → A[i, j])
end proc
```

```
> # For instance, we can obtain Eq. [13] of the manuscript as
  follows:
  Mat(1,1,kp1,km1,k2);
```

$$\begin{bmatrix} -k2 - km1 & 0 & kp1 \\ k2 & 0 & 0 \\ km1 & 0 & -kp1 \end{bmatrix} \tag{4}$$

```
> # This is for 3 enzymes and 7 substrates
  Mat(3,7,kp1,km1,k2);
```

$$\begin{bmatrix} [-3\,k2 - 3\,km1, 0, 0, 0, 5\,kp1, 0, 0, 0, 0, 0], \\ [3\,k2, -2\,k2 - 2\,km1, 0, 0, 0, 6\,kp1, 0, 0, 0, 0], \\ [0, 2\,k2, -k2 - km1, 0, 0, 0, 7\,kp1, 0, 0, 0], \\ [0, 0, k2, 0, 0, 0, 0, 0, 0, 0], \\ [3\,km1, 0, 0, 0, -2\,k2 - 2\,km1 - 5\,kp1, 0, 0, 12\,kp1, 0, 0], \\ [0, 2\,km1, 0, 0, 2\,k2, -k2 - km1 - 6\,kp1, 0, 0, 14\,kp1, 0], \\ [0, 0, km1, 0, 0, k2, -7\,kp1, 0, 0, 0], \\ [0, 0, 0, 0, 2\,km1, 0, 0, -k2 - km1 - 12\,kp1, 0, 21\,kp1], \\ [0, 0, 0, 0, 0, km1, 0, k2, -14\,kp1, 0], \\ [0, 0, 0, 0, 0, 0, 0, km1, 0, -21\,kp1]] \end{bmatrix} \tag{5}$$